SWYNEX INTERNSHIP PROGRAM

# Task 2: Exploratory Data Analysis & Insights

Comprehensive Guide to Analyzing Cleaned Data, Visualizing Trends, and Extracting Business Insights

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| **Prerequisite** | Task 1 Cleaned Dataset (Approved) | **Tools Allowed** | Python (Pandas, Seaborn, Matplotlib), SQL, or Excel |
| **Deliverable** | EDA Jupyter Notebook / Script + Visualization Charts + 5 Key Insights | **Submission** | Updated GitHub Repository & Portfolio Documentation |

## 1. Objective & Task Overview

Now that your raw dataset has been cleaned, filtered, and formatted in **Task 1**, your objective for **Task 2** is to conduct an in-depth **Exploratory Data Analysis (EDA)**. You will extract statistical measures, uncover distributions, identify hidden trends, and present five actionable findings accompanied by visual charts.

## 2. Step-by-Step EDA Execution Plan

| Phase | Key Analysis & Metrics | Expected Output |
|---|---|---|
| **1. Descriptive Statistics** | Calculate mean, median, standard deviation, quartiles, skewness, and min/max values for numeric metrics. | Summary statistics table (`df.describe()`). |
| **2. Univariate Analysis** | Examine individual variable distributions. Identify key categories, frequency counts, and variance. | Histograms, bar graphs, and frequency charts. |
| **3. Bivariate / Multivariate** | Analyze relationships between pairs of variables (e.g., Sales vs. Region, Churn vs. Customer Tenure). | Scatter plots, heatmaps, box plots, grouped bars. |
| **4. Pattern & Anomaly Detection** | Spot seasonality, growth trends, clustering, and unexpected outliers or anomalies. | Line graphs (time series) and residual plots. |

## 3. Requirement: Five (5) Actionable Insights

### Structure of a Business Insight

Each insight must go beyond describing data — explain the **What** (observed fact), the **Why** (underlying cause/trend), and the **Business Impact** (recommendation or strategic decision).

- **Insight 1 (Trend/Growth):** Identify significant overall growth, decline, or temporal patterns.
- **Insight 2 (Segment Performance):** Highlight top-performing and underperforming categories or groups.
- **Insight 3 (Correlation/Relationship):** Explain key dependencies found between variables.
- **Insight 4 (Anomaly/Outlier):** Detail an unexpected finding, spike, drop, or unusual cluster.
- **Insight 5 (Strategic Recommendation):** Provide a concrete action based on data findings.

## 4. Final Submission Checklist

- ☐ Executable code file (`.ipynb`, `.py`, or `.sql` query script) committed to your GitHub repo.
- ☐ At least 4–5 clear, labeled charts (with axis titles, legends, and clean color palettes).
- ☐ Updated project `README.md` containing summary statistics, chart summaries, and the 5 insights.
- ☐ Verifiable link uploaded to the evaluation portal for final approval.